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SLOW DEPOLARIZING AND HYPERPOLARIZING CURRENTS WHICH MEDIATE BURSTING IN *APLYSIA* NEURONE R15

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SUMMARY

1. Interruption of normal bursting activity by application of a voltage clamp reveals that action potentials in *Aplysia* neurone R15 are followed by two slow currents that long outlast the currents produced during the action potentials. Similar currents are seen following simulation of an action potential with a brief depolarizing pulse delivered under continuous voltage clamp.

2. One of these currents, herein called I_D, is an inward, or depolarizing current 0·5–5 nA in amplitude that reaches a peak 300–500 ms after the action potential. It produces the depolarizing after-potential that follows action potentials in this cell and is responsible also for the grouping together of action potentials into bursts.

3. The second current, herein called I_H, is an outward, or hyperpolarizing current 0·1–2 nA in amplitude that reaches a peak in 2–10 s and is still present for many tens of seconds following the action potential. I_H mediates the interburst hyperpolarization.

4. Both currents summate temporally during the burst. Despite changes in the amplitude and duration of action potentials during the burst, each action potential adds nearly constant increments to the summated amplitudes of I_D and I_H.

5. The summated amplitude of I_D grows during the first few action potentials and gives rise to the increased rate of depolarization and the increased firing rate seen during the first half of the burst. Due to its slower kinetics, I_H summates throughout the burst until its summated amplitude is large enough to cause the cell to hyerpolarize, thereby bringing the burst to an end.

6. When the normal burst is interrupted by application of the voltage clamp, the I_D and I_H current peaks are followed by a current which approaches a more negative steady-state level with a time course that consists of at least two phases. The first phase is exponential with a time constant of 15–30 s. Under continuous voltage clamp, the current following a train of depolarizing pulses returns to the holding current with a similar time course. These observations, together with time constants for I_H that are longer than the interburst interval, suggest that I_H is always partially activated during normal bursting.

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7. A computer simulation demonstrates that opposing inward and outward currents with different kinetics, i.e. I_D and I_H, are sufficient to give rise to bursting activity, in the absence of non-linear voltage-dependent conductances. Such voltage-dependent conductances, which are present in the normal cell, contribute to but are not necessary for bursting activity.

INTRODUCTION

Aplysia neurone R15 (Frazier, Kandel, Kupfermann, Waziri & Coggeshall, 1967) is a bursting cell. It exhibits a regular oscillation in its electrical activity that consists of 'bursts' of action potentials separated by interburst hyperpolarizations (Strumwasser, 1967; Frazier *et al.* 1967). Although the burst rhythm can be modulated by synaptic stimulation (Parnas & Strumwasser, 1974; Adams, Parnas & Levitan, 1980) or application of neurotransmitters (Wilson & Wachtel, 1978; Drummond, Benson & Levitan, 1980), it appears to be generated endogenously (Alving, 1968) through the activity of voltage-dependent ion channels in R15's membrane.

The ionic currents that contribute to bursting in neurone R15 have been investigated in a number of laboratories (for summary see Kandel, 1976). One of these currents is a steady-state (non-inactivating) inward current carried by Na^+ (Barker & Smith, 1978) or Ca^{2+} (Gorman, Hermann & Thomas, 1981). Different types of experiments have provided evidence for each of these ions as the charge carrier, and it may be that this current is in fact carried by both Na^+ and Ca^{2+} (Partridge, Thompson, Smith & Connor, 1979). The steady-state inward current is activated by depolarization, first appearing between -70 and -50 mV, and produces the characteristic 'negative resistance' region of the steady-state current–voltage (I–V) curve of the cell (Wilson & Wachtel, 1974). Treatments that eliminate the negative resistance region in a clamped cell (cooling, bath application of dopamine, synaptic input), presumably by reduction of this current, also produce a cessation of bursting in unclamped cells (Wilson & Wachtel, 1974, 1978; Adams *et al.* 1980). However, a negative resistance region, which appears to be produced by a similar current, has also been observed in non-bursting cells (Partridge *et al.* 1979). Thus a steady-state inward current, and the resulting negative resistance region in the I–V curve, are not by themselves sufficient to produce bursting, but may provide a continual depolarizing drive that takes the membrane potential to the action potential threshold, and thereby initiates the burst.

If the steady-state inward current provides a continual depolarization to the cell, then there must exist a second current which is activated by some aspect of the cell's activity during the burst, and which finally causes the termination of the burst. One candidate for this role is the Ca^{2+}-activated K^+ current which has been described previously in R15 (Meech, 1972; Gorman & Hermann, 1979) and other cells. Supportive evidence for this hypothesis comes from the demonstration, by use of intracellular aequorin or Arsenazo III, of an influx of Ca^{2+} with each action potential in the burst, and an intracellular accumulation of Ca^{2+} throughout the burst (Stinnakre & Tauc, 1973; Gorman & Thomas, 1978). However, as attractive as this hypothesis may be, we demonstrate in the accompanying paper (Adams & Levitan, 1985) that the transient slow outward current that is elicited by action potentials

in R15, and is responsible for terminating the burst, is not carried by K^+. A similar conclusion has been drawn by Kramer & Zucker (1985*a*, *b*) for the left upper quadrant cells of the *Aplysia* abdominal ganglion. Thus, another charge carrier must be proposed for the slow transient outward current responsible for terminating the burst.

A third current that appears to contribute to bursting is a transient inward current, with time constants of several hundred milliseconds, that is also activated by action potentials. This inward current has been observed as a depolarizing after-potential that follows each action potential in freely bursting cells (Thompson & Smith, 1976); it appears to cause action potentials to be grouped together into bursts. As such, it may be considered that it is the presence or absence of this transient inward current that distinguishes bursting from beating cells. Despite its similar role in different bursting cells, recent results (Kramer & Zucker, 1985*a*) suggest that the mechanism by which this inward current is produced may differ in different bursting cells from the same animal.

In this paper I describe the kinetics of the transient but long-lasting inward and outward currents in neurone R15, activated either by naturally occurring action potentials or by brief depolarizing pulses (simulated action potentials) presented under voltage clamp. In addition, I present a computer simulation which demonstrates that the difference in kinetics of these two action-potential-activated currents can give rise to bursting activity, even in the absence of other voltage-dependent conductances. The accompanying paper describes the voltage dependence and the ion dependence of the two transient currents (Adams & Levitan, 1985). Portions of this work have previously been reported in abstract form (Adams & Levitan, 1978, 1981, 1982).

METHODS

Animals

Adult specimens of *Aplysia californica* weighing between 100 and 300 g were obtained from Marine Specimens Unlimited, Pacific Palisades, CA, U.S.A. They were maintained at 18–20 °C in circulating artificial sea water (Instant Ocean) in 600 l tanks for up to 4 weeks before use.

Preparation

The abdominal ganglion was removed through a longitudinal incision in the ventral foot and pinned out on Sylgard (Dow–Corning) in a 25 mm plastic Petri dish. During the pinning, the sheath of the ganglion was stretched to facilitate penetration with micro-electrodes. Normal *Aplysia* medium (in mM: Na^+, 460; K^+, 10; Ca^{2+}, 11; Mg^{2+}, 55; Cl^-, 602; Tris or HEPES buffer, 10; glucose 10 g/l; pH 7·4) was continually circulated across the ganglion at a rate which would provide one complete change of medium every 2 min. Neurone R15 (Frazier *et al.* 1967) was identified by its size, location, white pigmentation and, after electrodes were inserted, by its electrical activity.

Recordings

Glass micro-electrodes were pulled from 1 mm o.d., 0·5 mm i.d. Pyrex capillary tubing containing a glass filament and filled with 3 M-KCl. Their resistance in *Aplysia* medium was 10–20 MΩ and dropped to 5–10 MΩ after penetration of the sheath. Standard electronic instrumentation was used and all experiments were recorded on an FM tape recorder for subsequent analysis.

A grounded aluminium foil 'flag' was placed between the electrodes to reduce capacitive coupling and the membrane potential rise time was generally less than 1 ms. The currents were filtered with a second-order filter with a time constant of 2 or 5 ms. With later experiments a sample–hold amplifier was inserted between the current signal and the filter and the amplifier was switched to the 'hold' mode during the depolarizing pulse.

RESULTS

Normally bursting cells

These experiments began with the observation that injection of a sinusoidal current into a non-bursting cell gave rise to burst-like activity, and prompted the question: is it possible to measure directly an oscillating current that is generated endogenously in R15? To accomplish this the cell was allowed to burst freely, and then a voltage clamp was turned on at various points during the burst cycle. In practice, the voltage-clamp holding potential was set to a particular voltage and then the clamp was turned on automatically when the membrane potential was changing with positive slope and became equal to the pre-set holding potential. This procedure avoided the capacitive transients that would otherwise be introduced at the beginning of the clamp by a rapid change in membrane potential and allowed measurement, almost instantaneously, of the current flowing through the membrane at the time the clamp was activated. Typical recordings of membrane current are illustrated in Fig. 1. When the clamp was activated during the depolarization between action potentials in the burst (left side), the current was inward (negative, or depolarizing), quickly reaching a peak and then relaxing somewhat. The time-to-peak in this example reflects, at least in part, the time constants used to filter the current. Upon release of the clamp after 2·1 s, firing began anew. When the clamp was activated during the beginning of the interburst hyperpolarization (Fig. 1, right side), the initial current was outward, and during the 2·1 s clamp period, changed only little. When the clamp was released, the interburst hyperpolarization phase continued. Later during the interburst hyperpolarization, when the cell was depolarizing, the initial current was inward and grew larger during the clamp period (Fig. 1, right).

The current measured shortly after application of the clamp varied cyclically during the burst cycle, as is illustrated in the lower half of Fig. 2. The upper trace in Fig. 2 is a sample burst, shown to illustrate the times when the clamp was activated. The burst pattern in a given cell remains remarkably constant for long periods of time, and thus current measurements from different bursts can be compared. The current measurements, plotted in the lower curve, were made at the negative peak of the current or, in the absence of a peak, at 400 ms following activation of the clamp. During the depolarizing phase of the interburst, the currents were small but inward, i.e. depolarizing. The acceleration into the burst was accompanied by a rapid increase in inward current. During the burst, the inward current increased following the first few action potentials. Since the rate of change of membrane voltage (dV/dt) is proportional to the net current flow ($dV/dt = -i_{net}/C_m$; C_m = membrane capacitance), the increase in inward current flow will cause the rate of depolarization between action potentials to increase and the interval between them to shorten. The current and action potential firing frequency reached peaks near the centre of the burst and then decreased. A burst in R15 typically ends with a depolarizing after-potential that does not reach threshold for eliciting an action potential. At the peak of the depolarizing after-potential the current switches polarity to become outward, or hyperpolarizing, and remains outward during the hyperpolarizing phase of the interburst.

If the voltage clamp is activated during the burst and left on for a longer time,

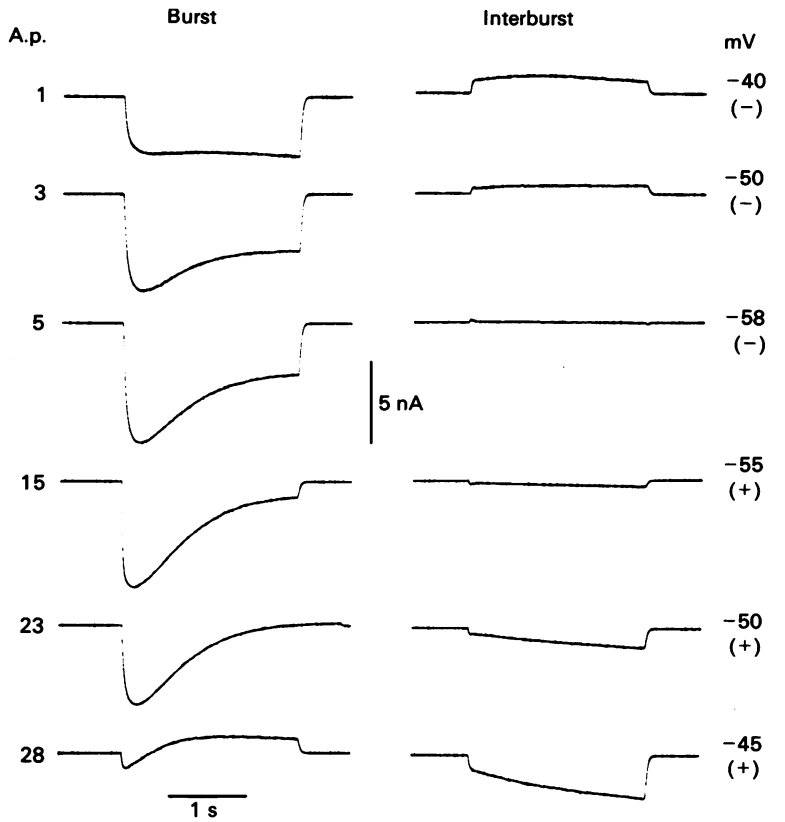


Fig. 1. Time courses of currents recorded after activating a voltage clamp at different phases during the burst. The cell was allowed to burst freely until the bursting pattern had stabilized. The clamp was then activated automatically for 2·1 s when the membrane potential became equal to the pre-set holding potential. The traces are arranged chronologically from the beginning to the end of the burst (left column, top to bottom) and from the beginning to the end of the interburst (right column, top to bottom). During the burst, the holding potential was set at −45 mV and the clamp was activated after each action potential as the cell was depolarizing. The numbers at the left of each trace refer to the number of action potentials (a.p.) that were allowed to occur during the burst before activation of the clamp. During the interburst period, the holding potential was varied to trace out the membrane potential excursions during this phase of the burst cycle. The numbers to the right of each trace indicate the holding potential and whether the cell was hyperpolarizing (−) or depolarizing (+) when the clamp was activated.

three distinct phases can be distinguished in the time course of the current flow (Fig. 3). Initially the current is inward, and it reaches a negative peak. The slope of the current then reverses and 2–10 s later a distinct positive peak is seen, although the net current at the positive peak may still be negative (Fig. 3*B*). The current then reverses direction once again and over a period of several minutes becomes increasingly large and negative. For want of a better term, I call this last period the 'recovery' phase. The currents reached during the recovery phase are always more negative than any seen during the earlier phases. When the clamp is turned on during the interburst period only the recovery phase is seen, beginning either immediately or after a few

seconds during which the value of the current remains relatively constant (cf. Fig. 1). The absence of negative and positive peaks during the interburst period demonstrates that the current peaks seen during the burst are not simply an artifact of the experimental procedure but, instead, arise from the activity of the cell.

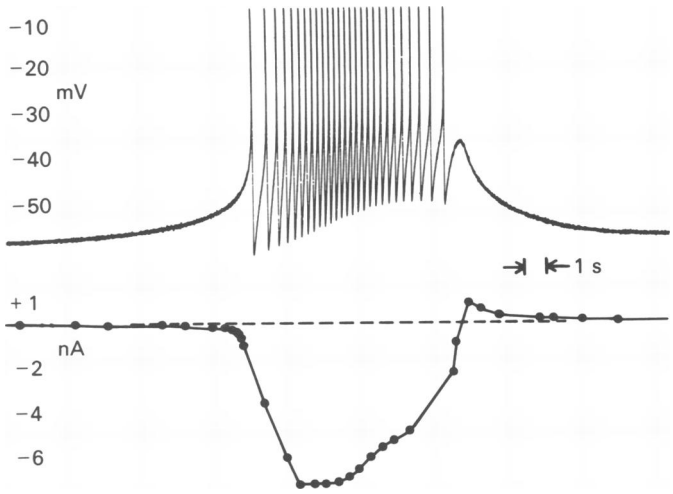


Fig. 2. Peak current shortly after activation of the voltage clamp. The upper trace is a sample burst to illustrate the times at which the clamp was activated. As in Fig. 1, the clamp potential was −45 mV for clamps applied during the burst and was adjusted to match the membrane potential for clamps applied during the interburst. During the burst, where a clear peak was visible in the current trace between 200 and 500 ms (Fig. 1, left column), the current was measured at the peak. During the interburst, the current was measured 400 ms following activation of the clamp. The dashed line indicates zero current.

In order to avoid confusion when comparing the currents described above with those measured when the cell remains voltage clamped, it may be useful to point out that differences in procedures give rise to different definitions of the 'zero current value'. The currents described in Figs. 1–5 come from cells that are bursting normally until the voltage clamp is turned on. The zero current value is true zero; no exogenous current is applied before the clamp is turned on, and the current traces begin at true zero (although current flow due to leakage around the electrodes has not been taken into account). When the clamp is left on for a long period of time (e.g. several minutes) the current during the recovery phase reaches a steady-state value of −10 to −20 nA. This steady-state current is equivalent to the holding current when the cell is held continuously voltage clamped, as in the experiments reported in Figs. 6–8 of this paper, Figs. 2–16 of the following paper (Adams & Levitan, 1985), and in the experiments described by Kramer & Zucker (1985*a*, *b*) in their measurements of two similar currents (their phase II and phase III currents) in *Aplysia* left upper quadrant cells.

A superposition of the currents that were measured by turning on the voltage clamp at different points during the burst is shown in Fig. 3*B* and a plot of the currents at the negative and positive peaks is given in Fig. 4*A*. The variation in the amplitudes

of the negative peaks was similar to that illustrated in Fig. 2, as the peaks reached increasingly larger negative values following the first three action potentials in the burst and became smaller, but still negative with subsequent action potentials. Following the first four action potentials, the current at the positive peaks was still

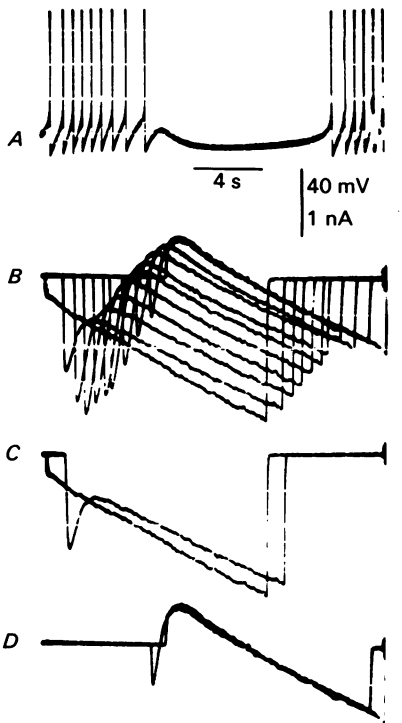


Fig. 3. Currents flowing during voltage clamps applied at various times during the burst cycle. *A*, sample voltage recording. *B*, superposition of current traces from voltage clamps applied just before the beginning of a burst, after each action potential, and after the depolarizing after-potential. Comparison of the current traces (*C*) before and after the first action potential and (*D*) before and after the depolarizing after-potential. The holding potential in each case was −45 mV. All oscilloscope traces were triggered just before the start of the burst when the membrane potential was depolarizing and reached −48 mV.

in the inward direction. However, after the fifth and subsequent action potentials the positive peaks became outward or hyperpolarizing. It may seem that, since the current following the fifth action potential became positive, the burst should have ended at that point, but closer examination reveals that threshold was reached for the sixth action potential before the current following the fifth became positive.

In analysing the above data, two points became clear. First, a normally bursting cell is never in a 'steady-state' condition. The recovery phase of the current, discussed further below, lasts several minutes, far longer than the time between bursts. Thus the currents are always changing slowly with time and any measurement of particular current components must take this into consideration. Secondly, a single action potential can elicit a large change in the slow currents. This second point is perhaps best illustrated by examining the individual pairs of current traces in Fig. 3*C* and *D*.

In Fig. 3*C* are the currents measured by switching on the voltage clamp just before and just after the first action potential in a burst. All of the oscilloscope traces were triggered when the membrane potential reached −48 mV just before the start of the burst. Thus the two current traces are positioned in 'real time' with respect to the voltage recording in the top trace, so as to account for the time-dependent changes in current. The difference between the two traces thus represents the effects of allowing a single action potential to occur *versus* holding the cell for a similar period of time at −45 mV. The current following a single action potential at first reached a peak more negative than the current measured in its absence. It then became slightly more positive and remained so for the duration of the clamp. In contrast to the difference in current wave forms measured before and after an action potential, the currents measured before and after the depolarizing after-potential coincided with one another (Fig. 3*D*), even though the delay between application of the voltage clamps was the same as that in Fig. 3*C*. The coincidence between traces in the absence of an action potential (Fig. 3*D*), and the difference between traces when an action potential is allowed to occur between them (Fig. 3*C*), argues strongly that the difference is in fact due to the occurrence of the action potential and suggests, moreover, that the difference between traces may provide a means of quantifying the effects of a single action potential on the slow currents.

The timing of the various phases of the current in Fig. 3 appeared to change somewhat during the burst, in part due to changes in the rate of depolarization between action potentials (cf. Fig. 3*A*) which led to changes in the time at which the clamp was activated. Since the data indicate that the negative and positive peaks represent current components that are activated by the action potentials that precede them, the reference chosen for the time-to-peak measurements was the time of the preceding action potential, rather than the time of application of the clamp. The delays between each action potential and the inward and outward current peaks that follow it are plotted in Fig. 4*B*. Both the inward and outward current peaks had a nearly constant delay from the action potentials that activated them. The inward current reached a peak at approximately 0·5 s following the action potential, with a slight increase in delay at the beginning and end of the burst. The outward current peak occurred 2·5 s after each action potential, with perhaps a slight decrease in the delay later in the burst.

The current traces in Fig. 3 were examined pair by pair in order to see if each action potential in a burst had an equal effect on the slow currents. The results are shown in Fig. 4*C* and *D* and are typical of all vigorously bursting cells examined. Each action potential in this cell elicited an inward current, measured as the difference between the current at the negative peak and the current from the voltage clamp preceding the action potential, of approximately 0·9 nA. In addition, each action potential elicited an outward current of approximately 0·25 nA. In twenty-four animals, the inward current per action potential was found to range from 0·5 to 3 nA, and the outward current per action potential from 0·1 to 0·5 nA. In cells that were beating, or were bursting slowly with little or no increase in firing rate at the centre of the burst, the inward currents were small, and variable in amplitude.

The largest currents are measured during the recovery phase. At the end of several minutes the current may reach values 10–15 nA more negative than when the clamp

is first turned on. Smith, Barker & Gainer (1975) have described the time course of the recovery as exponential with a time constant of about 16 s. The recovery phase in the present experiments was also exponential, with a similar time constant, at least for the first 20 s (Fig. 5). After that time, however, the time course deviated from exponentiality. Although small errors in measurement of the final value can make an exponential curve appear non-exponential, that does not seem to be the case here.

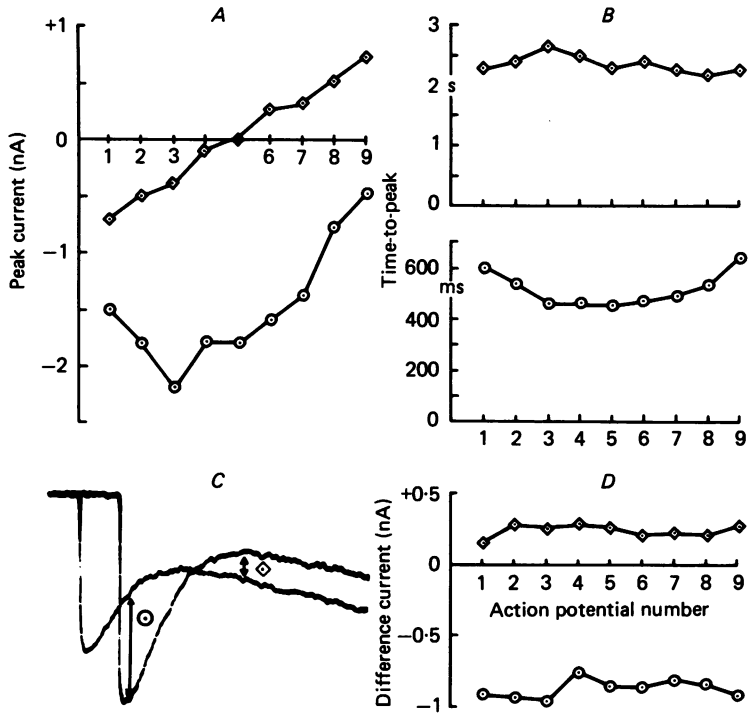


Fig. 4. Amplitudes and timing of the inward and outward currents. Data from Fig. 3. *A*, the current at the negative peak (⊙) was measured between 200 and 500 ms following application of the voltage clamp; that at the positive peak (◇) between 3 and 6 s. *B*, time-to-peak for the inward and outward currents. The times at which the negative peaks (⊙) and positive peaks (◇) occurred were measured from the time of occurrence of the action potential preceding the clamp. *C* and *D*, inward and outward difference currents elicited by each action potential in the burst. *C*, the current traces were compared pair by pair to measure the maximal inward difference current (left arrow) and maximal outward difference current (right arrow). *D*, the inward difference current (⊙) elicited by each action potential was *ca.* 0·9 nA and the outward difference current (◇) *ca.* 0·25 nA.

When the final value was perturbed by addition or subtraction of small amounts, and the resulting curves were replotted, they still deviated from exponentiality after no more than 30 s. In seven cells measured, the recovery seemed to consist of an initial exponential phase with a time constant between 15 and 30 s followed by a slower phase of recovery. The amount of recovery during the second phase was significant, but too small to test whether it was also exponential.

In a sense, the 'recovery phase' follows the transition of the cell from a normally

bursting state to the steady state that is imposed by the voltage clamp. During the normally bursting state, the currents measured at the beginning of a voltage clamp to −40 mV are always quite close to zero, at most 5 nA positive to 6 or 8 nA negative. These figures exclude, of course, the currents that flow during action potential generation, but otherwise are valid during any phase of the burst, including the interburst phase. By comparison, the steady-state current at −40 mV reaches some 10–20 nA negative. It thus appears that some conductance(s) remains at least partially activated (or inactivated) throughout the burst cycle, and that the current recorded during the recovery phase measures the deactivation (or reactivation) of this conductance.

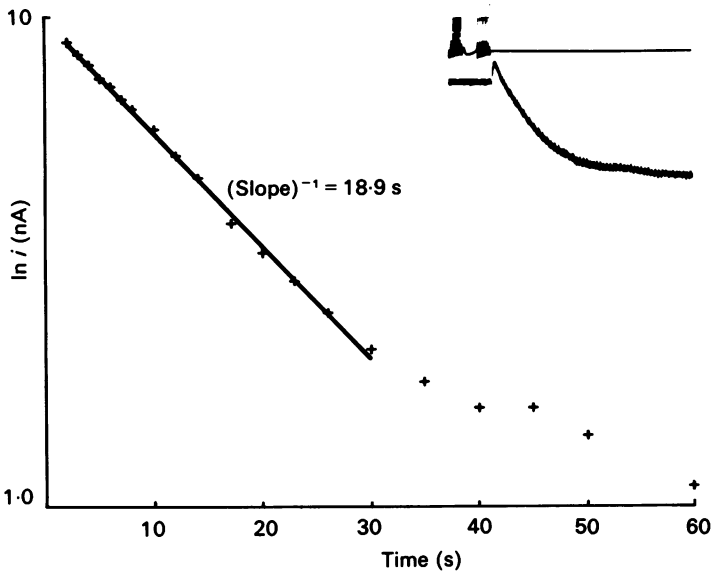


Fig. 5. Time course of the recovery phase of the current. The voltage clamp was activated just after the last action potential in a burst and held active for 200 s. The current values have been shifted to give a value of zero at the asymptote. Following the positive peak, the initial part of the recovery phase was well fitted by an exponential with time constant 18·9 s, but the initial exponential phase accounted for only about 75% of the recovery phase. Shifting the current trace up and down by small amounts before plotting on semilogarithmic coordinates did not change the two- (or more) component character of the resulting curve. Inset: the original voltage and current records from which the individual points of the semilogarithmic current plot were taken.

Simulated action potentials

The advantages of dealing with cells that are bursting normally are that the electrical activity is generated endogenously and that the slow currents measured in such experiments can be related directly to the pattern of burst activity. The main drawback is that the cell is never in a 'steady state'. As a result, all currents must be measured against a constantly changing background. In addition, any change in the cell's activity, whether or not it is mediated directly by the burst currents, may produce changes in the measured slow currents. In an attempt to circumvent these shortcomings, I examined the slow currents elicited by action potentials or by brief

depolarizing pulses in cells that were in steady state. This was accomplished in one of two ways. In both, the cell was held voltage clamped 10–20 mV below the action potential threshold (which was usually −20 to −30 mV) until the holding current had stabilized. The clamp was then either released briefly to allow the cell to fire an action potential, or stimulated with a 10 ms command pulse to depolarize the cell

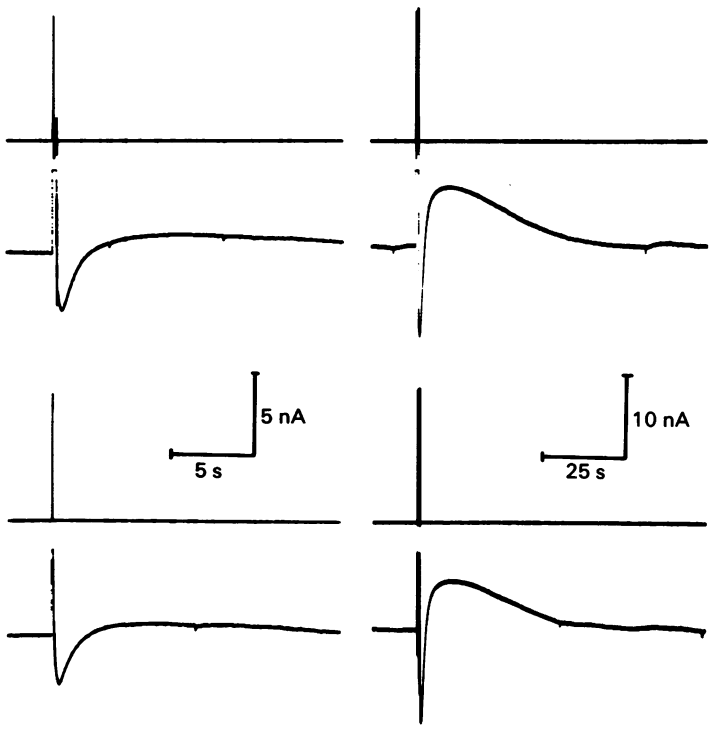


Fig. 6. Comparison of the slow currents elicited by action potentials and by depolarizing pulses. The cell was held voltage clamped at −40 mV until the holding current had stabilized. Top: the clamp was released to allow the cell to fire one (left) or five (right) action potentials and then the clamp was reactivated to measure the slow currents. Bottom: the cell remained clamped and the membrane potential was pulsed to +50 mV for 10 ms either once (left) or five times at 0·2 s intervals (right). Note the different time and current calibrations on the left and right side.

to +40 or +50 mV. The slow currents elicited by releasing the clamp and allowing either one or five action potentials to fire are shown in the upper two sets of traces, and those elicited by presenting one or five depolarizing pulses under voltage clamp are shown in the lower traces, of Fig. 6. The currents elicited with the two procedures were similar in both amplitude and kinetics. Single action potentials or depolarizing pulses (left side) elicited inward currents of 3–4 nA that reached peaks after 450 ms. Five action potentials or depolarizing pulses (right side) elicited a summated outward current of 6–7 nA that reached a peak after 10 s. In contrast to the endogenously firing cell, the outward current under these conditions was sometimes as large as or larger than the inward current, and the 'recovery phase' returned the current to the base-line holding current rather than to some current much more negative than the

starting point. As it proved easier to control stimulus parameters when the cell remained voltage clamped throughout the measurements, depolarizing pulses were used to simulate action potentials in all the following experiments.

The existence of a steady base line provided a more convenient method for measuring the slow currents than was available with endogenously bursting cells.

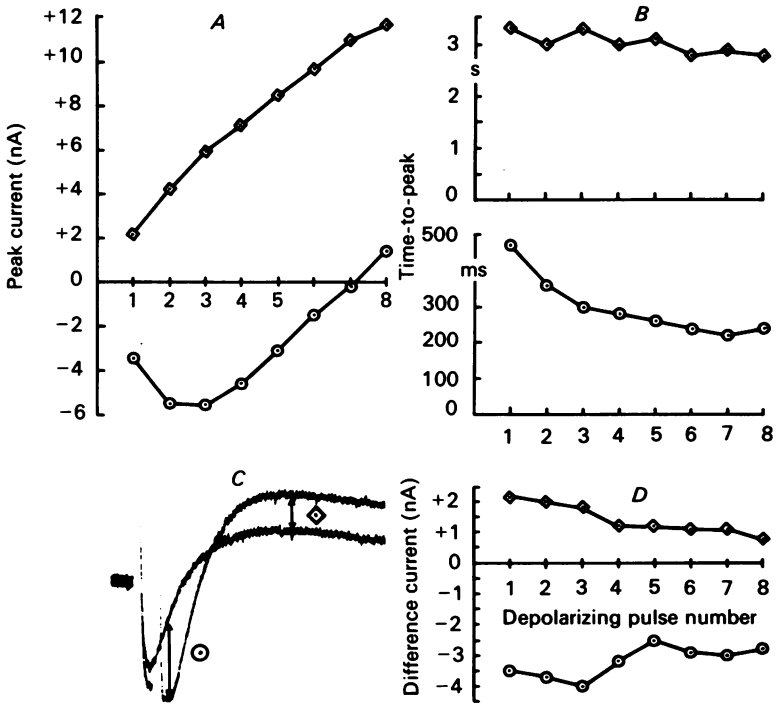


Fig. 7. Amplitudes and timing of the inward and outward currents elicited by a train of one to eight depolarizing pulses. A series of one to eight depolarizing pulses (holding potential −40 mV, pulse potential +50 mV, pulse duration 10 ms, pulse interval 0·5 s) were delivered under voltage clamp. *A*, the currents at the negative (⊙) and positive (◇) peaks were measured with respect to the holding current. *B*, time-to-peak for the inward and outward current. The times at which the negative peaks (⊙) and positive peaks (◇) occurred were measured from the time of delivery of the last depolarizing pulse. *C* and *D*, inward and outward difference currents elicited by each depolarizing pulse. *C*, the current traces were compared pair by pair to measure the maximal inward (left arrow) and outward (right arrow) difference currents. *D*, the maximal inward (⊙) and outward (◇) difference currents elicited by each pulse. Compare with Fig. 4.

Fig. 7*A* shows the amplitudes of the transient inward and outward currents that were elicited by trains of one to eight depolarizing pulses. This graph should be compared with Fig. 4*A*. Shortly following each depolarizing pulse, the current could be seen to reach a negative peak. These inward currents summated and increased in amplitude through the first three depolarizing pulses. With subsequent pulses, the inward current peaks reached less negative values, and after the seventh depolarizing pulse, the current measured at the negative-going peak took on a positive value with respect to the base-line holding current. The outward current peaks were always

positive and summated nearly linearly for up to eight depolarizing pulses. It should be borne in mind when comparing Figs. 4 and 7 that the zero-current reference for Fig. 4 is true zero and the zero-current reference for Fig. 7 is the base-line holding current that preceded generation of the pulses. As a result, the outward currents plotted in Fig. 7 were always positive and the inward current peaks could take on positive values.

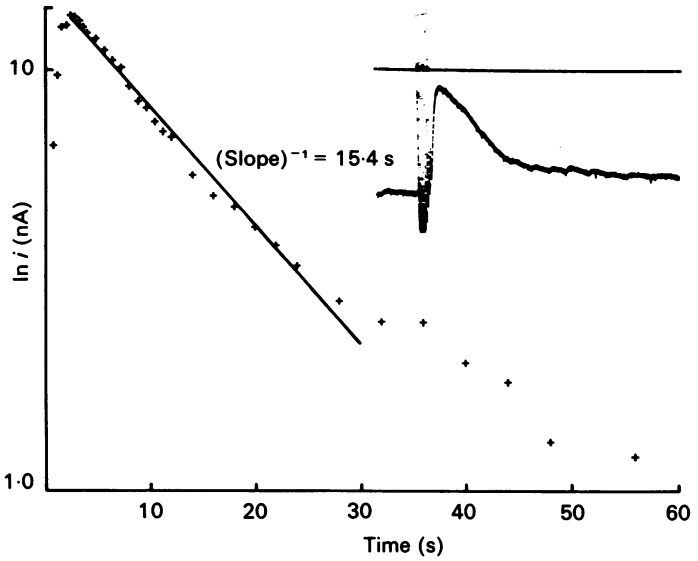


Fig. 8. Time course of the recovery phase following a train of eight depolarizing pulses. Following the positive peak, the current returns to the holding current with a time course similar to that of the 'recovery phase' in normally bursting cells (Fig. 5). The initial segment of the curve is well fitted by an exponential with time constant 15·4 s. Inset: voltage and current records.

The time-to-peak measurements, as a function of depolarizing pulse number are shown in Fig. 7*B* for comparison with Fig. 4*B*. Since the cell was continuously clamped and the depolarizing pulses were generated at regular intervals, the reference for timing in Fig. 7 was the time of occurrence of the preceding pulse. The peak of the inward current (bottom) following a single pulse occurred 470 ms after the pulse. Later in the train the peak was found closer and closer to the pulse. In contrast, the time between the depolarizing pulse and the peak of the outward current remained practically unchanged throughout the train of pulses (Fig. 7*B*, top).

The amplitudes of the slow currents elicited by each individual depolarizing pulse in a train of eight are shown in Fig. 7*C* and *D*. The procedure was identical with that used in Fig. 4, i.e. the currents attributed to the nth depolarizing pulse were obtained from the difference in the current after a train of n depolarizing pulses and the current after a train of $n-1$ pulses. In contrast to the results obtained with normally bursting cells (Fig. 4), where each action potential appeared to be equally effective in eliciting inward and outward slow currents, the outward currents elicited by depolarizing pulses at the end of a train were smaller than those elicited by the earlier pulses. There was a certain amount of variation in the inward current as well, but the change in

amplitude was less than 25 %. For the outward current, however, the current elicited by the eighth pulse in a train was less than half the size of that elicited by the first.

The time course of the recovery phase following a train of pulses (Fig. 8) resembled closely that found in normally bursting cells (Fig. 5). An initial exponential phase lasting 20–30 s was followed by a second slower phase of recovery. Time constants for the initial exponential phase were in the range 15–25 s.

Simulation

Incorporation of the currents described above into a simple electrical model of the cell (Fig. 9*A*) produced an oscillatory activity (Fig. 9*B*) that resembled, qualitatively at least, bursting behaviour in R15. For this simulation, 'action potentials' were produced by a threshold element that generated a 50 ms pulse when the membrane potential reached −30 mV. In order to provide a continual depolarization, the 'leak' conductance was chosen similar to the typical input resistance of R15 (20 MΩ). The leak conductance was independent of voltage, i.e. no attempt was made to simulate the negative-resistance region that is so prominent in the steady-state *I–V* curve of R15. The transient inward, or depolarizing slow current (labelled I_D) and the transient outward, or hyperpolarizing slow current (labelled I_H) were modelled by exponentially decaying current sources, independent of membrane potential.

The activity of the model (Fig. 9*B*) displays many of the features of bursting in R15 (cf. Fig. 2 for comparison). During the first part of the burst, the rate of depolarization and the action potential firing rate increase; later in the burst they again decrease. The burst ends with a characteristic depolarizing after-potential followed by a fairly rapid hyperpolarization that is due not to a sudden increase in hyperpolarizing current, but rather to the rapid decay of I_D. I_H remains partially activated throughout the burst cycle, and the small increases with each action potential during the burst are sufficient to eventually make the net current outward and to bring the burst to an end. Selection of parameter values is not especially critical, but changes in values give rise to predictable changes in burst pattern. Increases in the amplitude or decay time constant of I_D produce longer bursts with more rapid firing. Increases in the amplitude or decay time constant of I_H produce longer and deeper interburst periods. Omission from this model of either I_D or I_H leaves a circuit that is capable of producing beating activity but incapable of producing bursting.

DISCUSSION

I have described in this paper the kinetics of two slow currents in *Aplysia* neurone R15 that are activated by action potentials. Similar currents have recently been reported by Lewis (1984). It is important to note that these currents are measured as slow tail currents following naturally occurring or simulated action potentials, and are distinct from the currents underlying the action potentials themselves. One of these currents, I_D, is an inward, or depolarizing current that reaches a peak 300–500 ms after the action potential. I_D produces the depolarizing after-potential that follows action potentials in this cell and is responsible also for the grouping together of action potentials into bursts. Kramer & Zucker (1985*a*) have described a similar current (their phase II current) that seems to play much the same role in *Aplysia* neurones

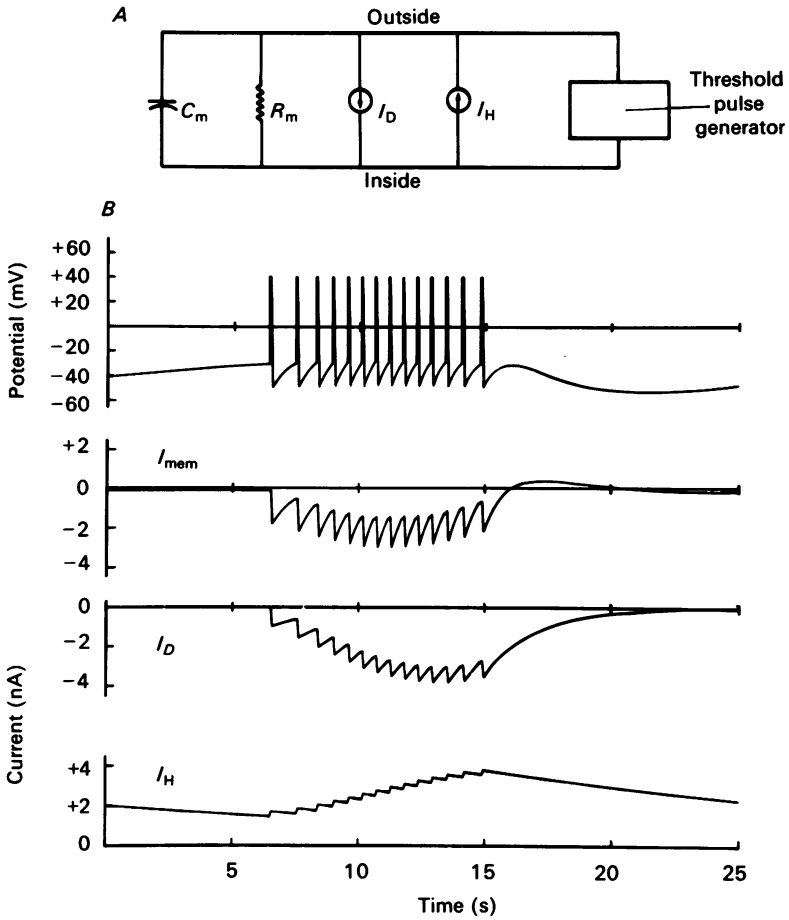


Fig. 9. *A*, simplified electrical model of the cell membrane for simulation of bursting. The model consists of the membrane capacitance (C_m), a leak resistance (R_m), a threshold element that, when the potential reaches −30 mV, generates a 50 ms pulse to +40 mV followed by a return to −50 mV, and two current generators that are incrementally increased by each 'action potential' and which then decay exponentially to zero. The inward current generator (I_D) is meant to simulate the transient inward current described in the previous Figures, and the outward current generator (I_H), the transient outward current. *B*, a simulated burst. The model parameters were: $C_m = 0{\cdot}05\ \mu F$; $R_m = 20\ M\Omega$; $I_D = 1$ nA/action potential, decay time constant = 2 s; $I_H = 0{\cdot}25$ nA/action potential, decay time constant = 20 s. The circuit equation describing the model circuit in *A* was integrated numerically (program available upon request). The traces below the voltage trace give the summated inward current (I_D), the summated outward current (I_H), and the total membrane current ($I_{mem} = I_D + I_H + I_{leak}$).

L2–L6. However, a comparison of the conclusions reached in the following paper (Adams & Levitan, 1985) with those reached by Kramer & Zucker (1985*a*) suggests that activation of I_D in neurone R15 and of the phase II current in neurones L2–L6 occurs through somewhat different mechanisms.

The second current, I_H, is an outward, or hyperpolarizing current that is smaller

and slower than I_D. I_H reaches a peak in 2–10 s and is still present for many tens of seconds following the action potential. Because of the long duration of I_H, the incremental contributions from each action potential in the burst summate, until the summated amplitude is large enough to cause the cell to hyperpolarize, thereby bringing the burst to an end. A similar current (the phase III current of Kramer & Zucker, 1985*b*) has been described for *Aplysia* neurones L2–L6.

One of the characteristics of bursting in R15 is that the duration of action potentials in the burst successively increases. This 'spike broadening' appears to result from inactivation of a K^+ conductance (Aldrich, Getting & Thompson, 1979) and can vary in extent from a fairly modest 10–20 % up to 100 % or more in some cells. In spite of this, there is very little variation in the amplitudes of I_D and I_H elicited by successive action potentials (Fig. 4). One possible explanation for the constancy of I_D and I_H in the face of the increase in action potential duration is provided by observations of Ca^{2+} influx during bursting (Smith & Zucker, 1980). Those experiments made use of intracellular injection of the Ca^{2+}-sensitive dye Arsenazo III and demonstrated a nearly equal influx of Ca^{2+} with each action potential. We will show in the accompanying paper that both I_D and I_H are Ca^{2+} dependent, so that the constancy of their amplitudes is consistent with the possibility that they are triggered by the influx of Ca^{2+} during the action potentials.

Both I_D and I_H can be elicited by naturally occurring action potentials, by single action potentials that occur when a voltage clamp is released for a short period of time, or by brief depolarizing pulses presented under voltage clamp to simulate action potentials. In all three cases, the amplitudes of I_D are similar. The amplitude of I_H, however, depends on the method used to elicit it. When I_H is measured by voltage clamping a freely bursting cell, its amplitude is always smaller than that of I_D. When the cell is held voltage clamped to prevent activity, and I_H is elicited either by releasing the clamp to allow an action potential to occur or by simulating an action potential with a depolarizing pulse, then its amplitude can be as large as, or larger than that of I_D. This observation suggests some sort of activity-dependent inactivation of I_H, but not of I_D. Taking into consideration the fact that the duration of I_H is much longer than the interval between action potentials in a burst, and as long as, or longer than the interval between bursts, it may be that I_H is always partially activated in a normally bursting cell, and that the incremental activation by subsequent action potentials is not as effective as when I_H is activated in a cell at rest. This idea is supported by the decreasing amplitude of I_H elicited by later depolarizing pulses in a train (Fig. 7*D*), and the somewhat less-than-linear summation of I_H that was illustrated in Fig. 7*A*. In contradiction, however, is the observation that each action potential elicits an equal amplitude I_H in a normally bursting cell (Fig. 4*D*). The cause for this discrepancy is not clear, unless it is simply that I_H is already so activated in a normally bursting cell that it is little affected by the further activation from individual action potentials.

A sustained partial activation of I_H throughout the burst cycle would account also for the large currents seen during the recovery phase in a normally bursting cell. Such a sustained activation appeared in the computer simulation of bursting (Fig. 9) as a consequence of the kinetics of I_D and I_H, rather than as a specific attempt to include it. The similarity between the time course of the recovery phase in a normally bursting

cell (Fig. 5) and the time course of the decay of I_H elicited by a train of depolarizing pulses (Fig. 8), also suggests that the current recorded during the recovery phase measures the slow deactivation of I_H.

In the experiments reported in this and the accompanying paper, recordings were made from cells that remained attached to their axons. Thus, the current measurements obtained by voltage clamping the soma reflect, to some extent, currents that arise from imperfectly clamped regions of the axon. Although these currents complicate interpretation of the data, the accompanying paper (Adams & Levitan, 1985) demonstrates that some of them seem to be necessary for generation of bursting activity in R15. Treistman (1980) has also reported that the axon contributes to pace-maker activity. Thus it appeared necessary to leave the axons attached in order to study the mechanisms of normal bursting in R15. On the other hand, it has been reported (Alving, 1968; Chen, von Baumgarten & Takeda, 1971; Schacher, Proshansky & Camardo, 1982) that restoration of bursting sometimes occurs when cells are axotomized, removed from the ganglion, and placed in cell culture. It is not yet clear, however, how the properties of cultured cells change so as to allow for this restoration of activity, and thus it remains to be determined whether bursting in such cultured neurones is mediated by the burst currents described here.

The simulation of bursting behaviour that was presented in Fig. 9 was not intended to incorporate all known information about the currents in R15, but rather to demonstrate that no special I–V relationships are necessary to produce rhythmic activity. All that is required is a steady depolarizing drive to get the burst started (this is provided in the simulation by the leak conductance), a threshold element to produce pulses, a brief depolarizing after-current (I_D) to cause the action potentials to group together into bursts, and a slow hyperpolarizing after-current (I_H) that summates during the burst and finally causes the burst to end. In particular, there is no need for a negative resistance element or a region of negative slope resistance in the cell's I–V curve. To be sure, such features must affect the exact form of the bursts in a normally bursting cell, but they are not necessary for burst production.

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